HUMANOID

A Final Year Project Report submitted to Manipal Academy of Higher Education in partial fulfilment of the requirement for the award of the degree of

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In

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ABSTRACT

Humans could be the most advanced futuristic creatures of nature till now. The main purpose building and manufacturing machines or robots is to reduce the physical and mental repetition of the work, which would in the time help in stress and provide relief and comfort to humans till the end of the time.They also let us do the necessary and more important things in life than just do the preliminary work, present themselves where humans cannot, like space, sun, or hidden caves, or view in the dark, examine the situation etc.

As mentioned above the ultimate goal is to improve nature's ultimate invention to be stronger and sharper to do more productive things. The Humanoid here is modelled initially in the CAD file with taking all the necessary details, after testing the stress tensor in SOLIDWORKS. The 3D file was then imported in a STL format with a URDF XML file corresponding to the parts of the robot containing the dimensions and angles of the components. The imported file is run on a Neural Network for the simulation on Rviz. Multiple Neural Networks have been implemented and the proper weights have been determined. This is a state based algorithm rather than repeatedly checking on the weights of the Humanoid.

Considering the complexity and the error rate on the Neural Network mesh, the state based algorithm states the current state and repeats the weight every time it goes through the same values. THis concept is taken from the compression algorithm LZMA, helping the Neurons to lighter the weights regularly as the time goes, as the only weight here required is the state technically.

In conclusion, the humanoid presented here is more efficient than the 80% of the types in this world in imitating the human gait system. The algorithm basically compresses the data, trains it on the cloud, and projects the weights and values on the raspberry pi connecting the motors and links of the Humanoid.

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CHAPTER 1
INTRODUCTION

1.1

The intense development in the stream of Robotics and Artificial Intelligence has brought new changes to the scientific discipline. Introducing the idea of cloud computing, and the idea of how it is efficient and is better, the differences and other preliminaries of the old to new models and many more.

The so-called Machine Intelligence was founded back in 1956. It has always been a drag to do repetitive things, might be mental or physical work. it has always been a waste of time. Instead we could do a better job thinking about new ideas and creating new things. The so called robots were created to do those low jobs, which were not that efficient for industrial use. The humanoid has begun to build with a computer that controls and thinks, to make decisions about what to do. That is the motivation for this project, to build a GUI, based on machine learning algorithms and replicate that in real life hardware robots.

1.2

Machines like SOFIA and ATLAS have been built to be more human-like. they are stronger, harderr, and more able than us in everything. They have a rich personality and holistic cognitive AI. They can think at a supercomputer's speed. Create solutions humans could not ever think of.

Fig 1.1

The main motivation for the artificially intelligent humanoid robot is the thinking power. By the use of Google's reinforcement learning algorithm the robot is able to balance on its own. From the recent serverless technology of Amazon Web Services console compute engine, a strong computer with a powerful GPU machine could be accessed and the whole policy optimization for every joint on the humanoid with every combination with each other calculations is being done on that computer whilst the onboard computer is rather a mini small Raspberry pi 3b+, for signal processing and PID mostly.

$$\theta = \max \checkmark \; E[X \; H \; t{=}0 \; R(st)|\uparrow \checkmark]$$

1.3

The only drawback is the speed of the data received. The cloud compute engine chosen for this is a 16GB GPU. The objective being accomplished and processed on the cloud and the GUI created in JavaScript as it is a scripting language, it does not require compiling again and again. The humanoid needs to balance itself and walk in the simulation and must walk the same more or less.

Additionally the compression algorithm applied makes the data transfer easy and efficient, reducing the lag in the transmission, making the model more aware and real-time human impression.

The importance of this completion is this would be the first serverless cloud based data storage and compute engine based stable robot. The robot would be able to produce infinite data and process it on the automatic scalable instance. Google's algorithm would help in balancing the robot below the torso and manage the upper part by replicating the mean motion. the heavy electronics would not be needed, making the robot completely wireless and serverless.

CHAPTER 2
BACKGROUND THEORY

The forward and inverse kinematics, PID controllers, DNN, RL, and gait system analogy are the prerequisite things to be known. The humanoid robot is a great invention put up by all types of engineers and scientists to make a dream come true, to clone another human with greater capabilities in the physical and mental world. The miniature size and a GUI is produced in this paper.

2.1 Literature Review

	Paper	Author	Methodology	Review
1	Gait analysis for the development of the biped robot foot structure	Daichi MAITA	bipedal	to apply the human plantar arch to biped robots
2	Gait Analysis for a Human with a Robot Walking Helper	Sunil Agarwal	Bipedal but helper	to compute gait of humans walking with a robot helper.
3	Emergence of Locomotion Behaviours in Rich Environments	Nicolas heess	All Around bipedal environmental	learn to run, jump, crouch and turn as required
4	Deep Reinforcement Learning for Single-Shot Diagnosis and Adaptation in Damaged Robots	Harshita Nair	equipped	single policy is trained to adapt against a wide variety of damages
5	Policy gradient learning for a humanoid soccer robot	A. Cherubini	motion control	effectiveness of the policy gradient learning method
6	Evolving a Locus Based Gait for a Humanoid	Gordon Wyeth,	locus based	evolving a stable

	Robot	**Damien Kee**		**humanoid walking gait ihai is based around parameterised loci of motion.**
7	**An Evolutionary Gait Generator with Online Parameter Adjustment for Humanoid Robots**	**Amin Zamiri , Amir Farzad**	**learning speed is the highest**	**robustness of the developed walking behaviors even in extremely high and low speeds providing appropriate frequency**
8	**Genetic algorithm-based optimal bipedal walking gait synthesis considering tradeoff between stability margin and speed**	**Goswami dip**	**event based**	**The optimal walking gaits are experimentally realized on a biped robot**

2.2 Theoretical Discussions

A human-like robot mentioned and made in the above papers is the fact that making the face more rounded, the teeth like features. The features added to the robot from outside needs to be similar to the actual human being. the ear, the nose receptors for smell, the five senses acquired. The face cut of the robot if rounded makes the look more friendly, more approachable. The ergonomic sense of the robot which would make the person who is looking at it approachable, more interactive. Some things can also be added to the robot like IR sensors to the eyes, correcting the eyesight and broadening the light spectrum. Similar can be done with the ears. The electrosensitive material for skin is used for the touch feeling, sensitive to micron level. Generalizing and descending the values to be normal to the

macroscopic world making the robot most human like. A robot can only help a human if it is better than human in any aspect.

2.3 General Analysis

Our biped robot takes off two legs. The waist-link connecting the two legs, providing each leg has three links: foot-link, shank-link, and thigh-link at the end of the touch of the ground. The ankle is the joint between foot-link and shank link, the shank link and the thigh link is the knee, while the hip is the link between the main module and the thigh link.. The biped has 12 DOFs. Two DOFs at ankle, 1 DOF at knee, and 3 DOFs at hip, i.e., 6 DOFs in each leg. Each DOF is connected to an independent individual actuator.The links are made of Aluminium type 7800 quality, strong and light. Usually these are used in aircrafts The overall mass or inertia of the links are computed. The table formed gives us all the information from the python module from the code library, denoting the forward and inverse dynamics factors, that is, Denavit,Hartenberg Matrix. easily computable in python.
based on the positions and weights of the actuators which are located at the joints between two links. The link masses are assumed concentrated at the joints located at the distal
ends. The mass distribution of the biped is shown in Fig. 2. ‘‘d 1 ” indicates the thigh-link length and “d 2 ” the shank-link length. “F y ” and “F x ” are the foot width and length. “F x1 ” and “F x2 ” are the distances from the ankle to the rear and front edges of the foot. “B y ” and “B x ” are the hip-link width and length. The distance between two hip-joints is “w”, and The distance between the ankle-joint and CM of the foot is “L”.

2.4 Mathematical derivations

Algorithm 2 "Vanilla" policy gradient algorithm

Initialize policy parameter θ, baseline b
for iteration=$1, 2, \ldots$ **do**
  Collect a set of trajectories by executing the current policy
  At each timestep in each trajectory, compute
    the *return* $R_t = \sum_{t'=t}^{T-1} \gamma^{t'-t} r_{t'}$, and
    the *advantage estimate* $\hat{A}_t = R_t - b(s_t)$.
  Re-fit the baseline, by minimizing $\|b(s_t) - R_t\|^2$,
    summed over all trajectories and timesteps.
  Update the policy, using a policy gradient estimate $\hat{g}$,
    which is a sum of terms $\nabla_\theta \log \pi(a_t \mid s_t, \theta)\hat{A}_t$
end for

CHAPTER 3
METHODOLOGY

3.1

This methodology is divided into hardware and software, fabrication and GUI respectively. The fabrication is done by the 3D modelling in SOLIDWORKS and assembling the head, body, torso, arms and legs individually. An XML file was created to link and name the 3D files for importing into ROS for manipulating them individually. Once the files are imported and assembled into ROS, they are further opened in GAZEBO.

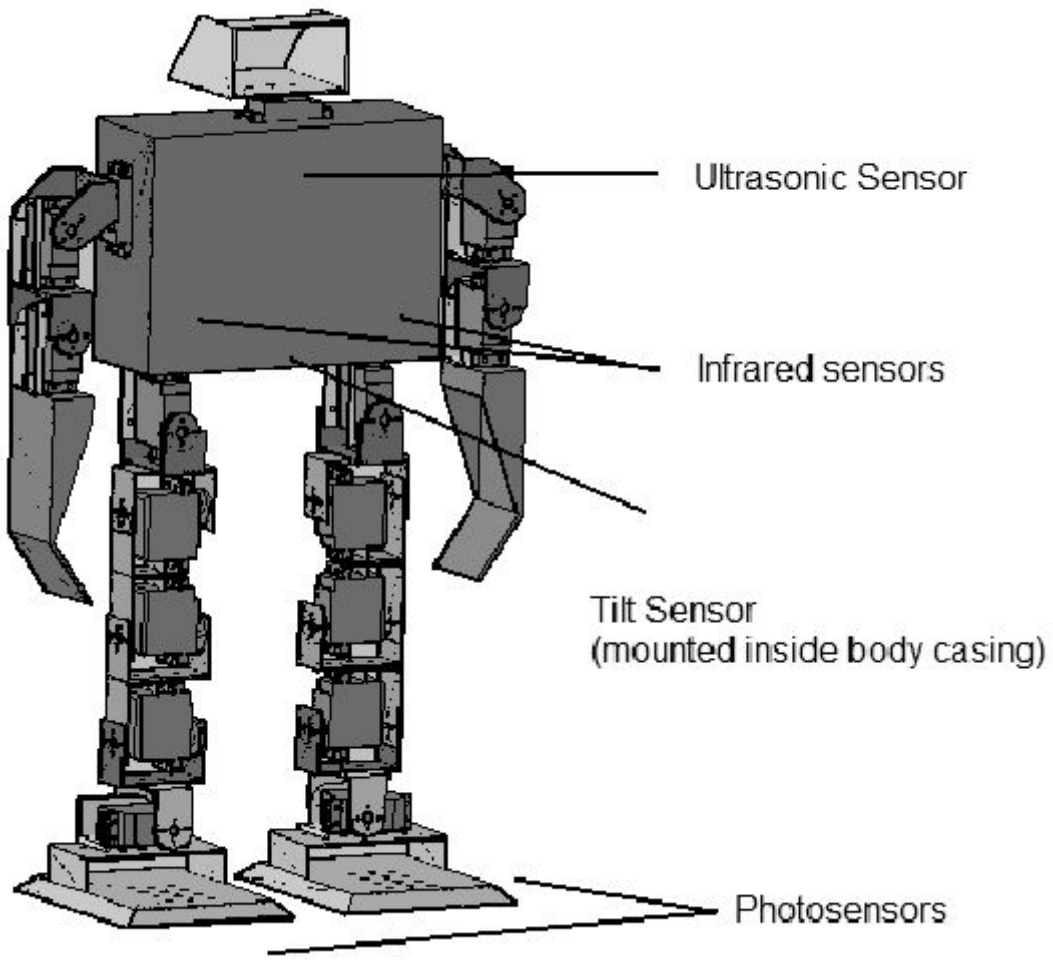


fig 2.1

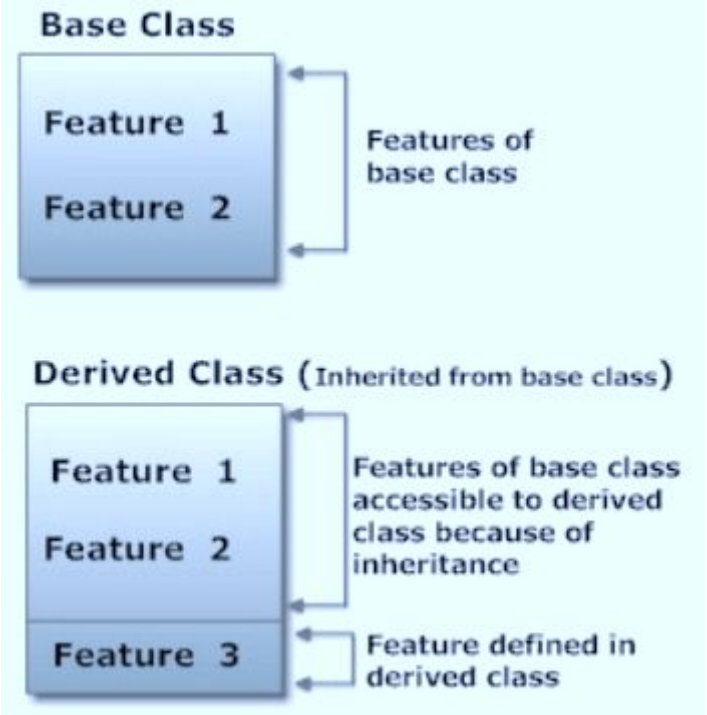
The Artificial intelligence and Automation code is written in python as it is an Object Oriented Programming. The special feature of Polymorphism, Encapsulation and Inheritance, and Classes and Objects make the code lesser complex i.e., The Big O notation O(n) = n^2 . That makes the simulation and the code simpler. The code is said to be non-repeatable, as that itself is the main goal of AI. The lesser repeatable it is, the less complex it becomes, and the

floating points are least. This is a simple OOP example of how the code connects with the 3D models and with each other and other files of DNN.

```
class Person: (a class is initiated)
    def __init__ (self, name, age): (a function inside a class)
        self. name = name
        self. age = age
    def show_salary (self): ( another function to call)
        print ("Salary is unknown")

class Employee (Person): (another class is initiated)
    def __ init__(self, name, age, salary): (another function)
        super ().__init__(name, age)
        self. salary = salary (another function)
    def show _salary(self):
        print ("Salary is", self.salary)

if __name__ == "__main__":
    p = Person("y", 23) (here the class is instantiated)
    x = Employee("x", 20, 100000)
    p.show_salary()  (functions are being called here)
    x.show_salary()  (and here)
```



The objects created in this humanoid are right arm, right leg, torso, and the head. The Classes created are one of DNN to control the integrated 3D model, Policy Optimization to control the limbs and arms including PID.

Applying the Markov's chain algorithm which efficiently powers up the entire silicon valley, we get the weights, processing them in the gradient descent on every node, we input them to the more stable weight to the motor after the PID.

Here is a small code of 2 Layer Neural Network:

```
def nonlin (x, deriv = False):  #sigmoid
   if(deriv == True):
      return  x* (1-x)the dataset is derived from the accelerometer
   return  1/(1 +np.exp(-x))
   X = np .array([  [0,0,1],      # dataset
            [0  ,1  ,1],
            [1  ,0  ,1], here  the dataset is being taken by the gyroscope
            [1  ,1  ,1] ])
y = np. array([[0,0,1,1]]).T      #output data
np .random.seed(1)
syn0 =  2*np.random.random((3,1)) - 1
for i ter in xrange(10000):
   l0  = X                        #forward propagating
   l1 =  nonlin(np.dot(l0,syn0))
   l1_ error = y - l1 the error is being propagated,
   l1_ delta = l1_error * nonlin(l1,True) so here it is being removed
   syn0 + = np.dot(l0.T,l1_delta)       #update
```

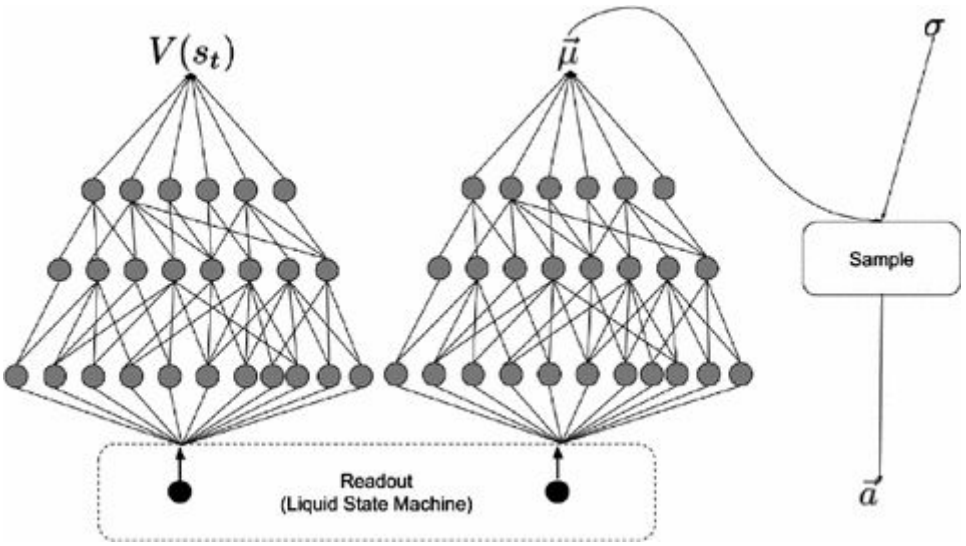
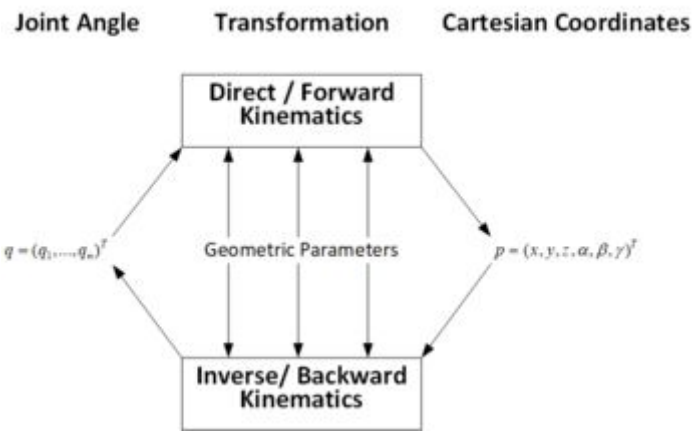


Fig 3.1

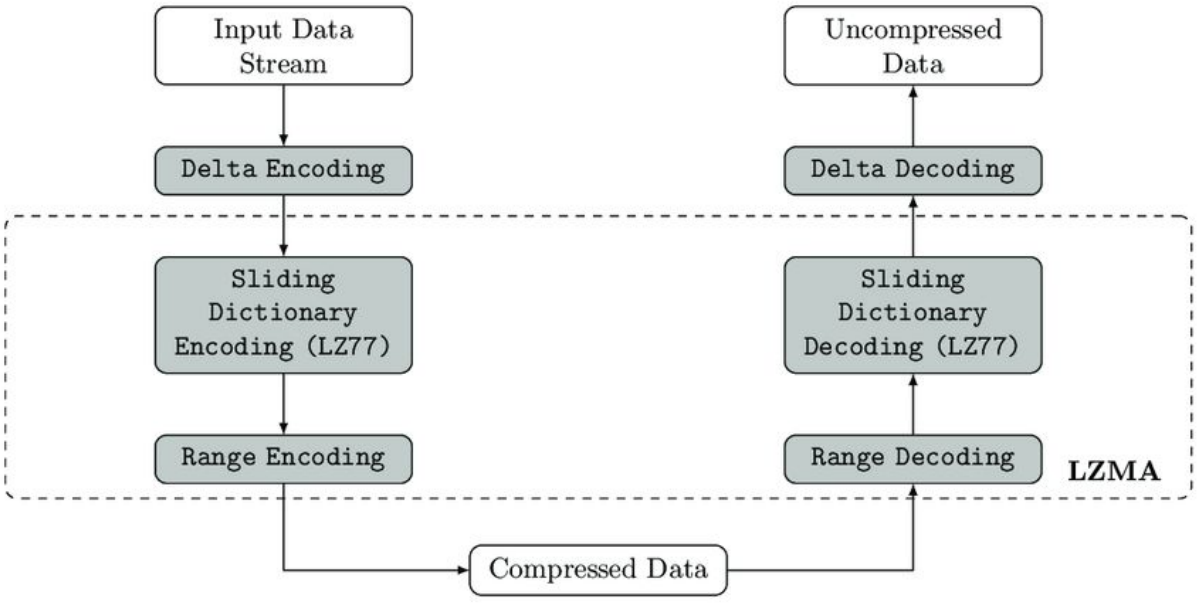
Sigmoid function is used as logarithmic regression is being used in the Neural Network. The V and Mu in the above figure shown are for limbs and stability.

Once the classes and Objects are set, the following is the diagram of a flowchart which shows the propagation of code and launch file flow. The launch file is also written in XML, as well as the POM.file which acquires depositories. Those depositories contain additional files when run adds missing files automatically. The gait parameters destined by the forward and inverse kinematics, by the Danavit, Hartenberg Matrix i.e.,

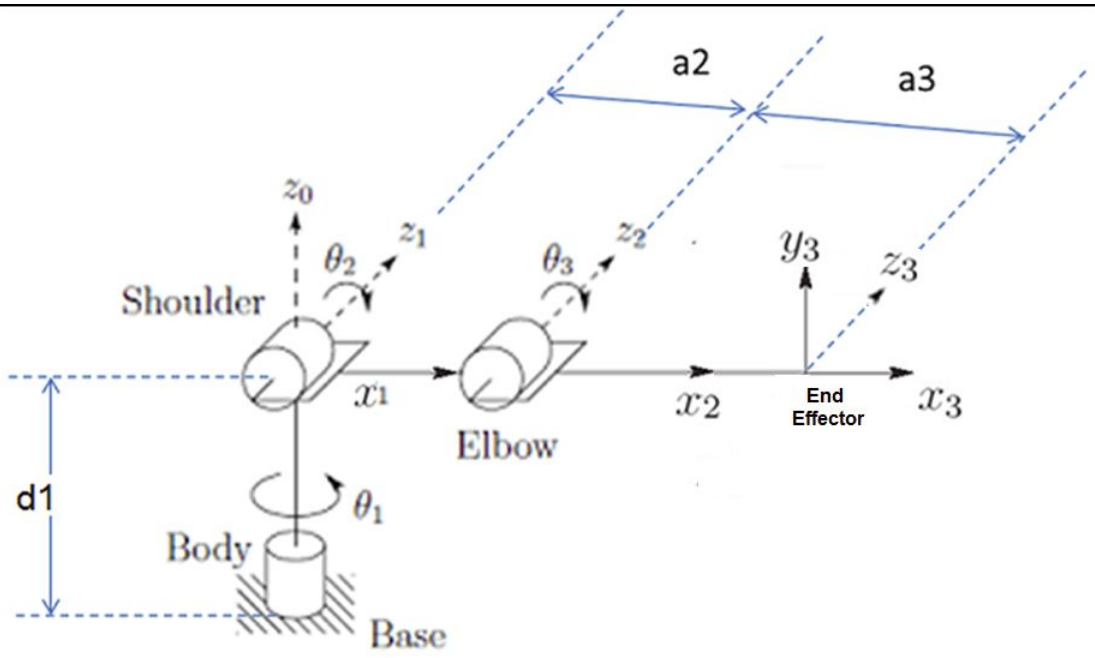
$$^{i-1}T_i = \begin{bmatrix} \cos\theta_i & -\sin\theta_i\cos\alpha_{i,i+1} & \sin\theta_i\sin\alpha_{i,i+1} & a_{i,i+1}\cos\theta_i \\ \sin\theta_i & \cos\theta_i\cos\alpha_{i,i+1} & -\cos\theta_i\sin\alpha_{i,i+1} & a_{i,i+1}\sin\theta_i \\ 0 & \sin\alpha_{i,i+1} & \cos\alpha_{i,i+1} & d_i \\ 0 & 0 & 0 & 1 \end{bmatrix},$$



Once the factors are determined for all the limbs and the body, we need to create a variable in the objects and direct the objects to the class with the factors for DNN and NN, which take the input factors of T, proceeds to forward propagation and then backward propagation, checks and updates the given weights, until the body stabilizes. To solve this complex network, we need a strong processor which would increase the weight of the Physical model, so the computing is done on the cloud services provided by AWS console Compute Engine. All the computations done on the cloud as the servers are controlled and depicted by AWS. The recent serverless technology gives us power to compute anywhere and use the data and GPU from any part of the world.
The fabrication part is done by a 3D printer. The factors taken into consideration were the strength required on all the limbs, the stress and strain developed by the motion/movement of the body parts. The 3D printed models are assembled to be a single part with the motors and sensors.



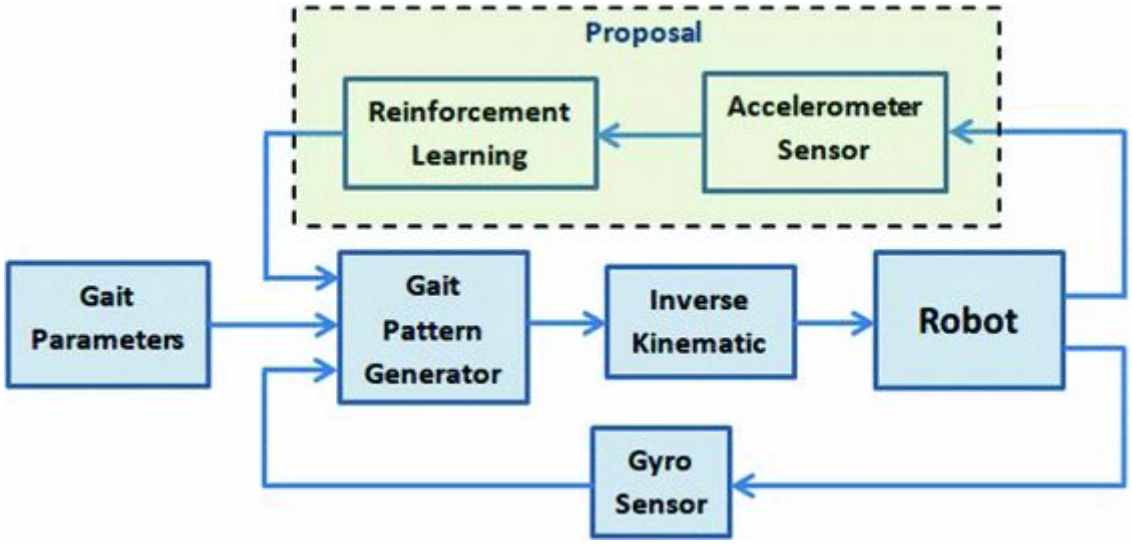
The motors used are servo motors customized to be a feedback 360 degree motors. The sensors used are pressure sensors, gyro and accelerometer sensors. The sensors are auto calibrated by the NN. The calibration needs to be done on the on board computer, as it needs to be quick. Error or lag cannot be appreciated here, it might cause failure. The On board computer chosen is Raspberry Pi 3b+ considering the cloud management and servers.
The Robot in hardware as a total has 24 motors, 2 gyro sensors, 2 pressure sensors, a Battery and a high current buck converter, and lastly an On board computer Raspberry pi 3b+ .



DH Parameter Convention for Three Links

Link	$\boldsymbol{\theta_i}$	$\boldsymbol{d_i}$ (mm)	$\boldsymbol{\alpha_i}$	$\boldsymbol{a_i}$ (mm)
1	$\boldsymbol{\theta_1}$	**133**	-pi/2	0
2	$\boldsymbol{\theta_2}$	0	0	**98**
3	$\boldsymbol{\theta_3}$	0	0	**220**

Fig 4.1



The Gait analysis is a very important and crucial factor in the humanoid as it determines the stability of the whole robot. The switching strategy is important to divide the walking cycle into several stages.

stage 1 : lift the left leg to its maximum for flexion and height

stage 2: lower the left leg until it makes complete contact with and the ground

stage 3: lift the right leg to its maximum for flexion and height

stage 4: lower the right leg until it makes one complete contact with the ground

stage 5: this stage follows the 1st or 3rd , and brings the one robot leg to a standing pose with both legs landed on the ground

Control Scheme	Online controller	Objective
Real time	Damping controlled (stage one and three, SSPs of stage two and four)	Eliminate the upper body oscillation in single support phase by imposed damping at the ankle joints
balance control	ZMP compensator (stage one and three, SSPs of stage two and four)	Maintain dynamic balanced by horizontal motion of the pelvis
	Soft landing controllers (DSPs of stage two and four)	Absorb landing impact and adapt the foot to the ground surface
Walking pattern	Pelvis swing amplitude controller (DSPs of stage two and four)	Compensate the lateral and swing amplitude of the pelvis by considering amplitude of the ZMP
control	Torso pitch/roll controller (DSPs of stage two and four)	Compensate the center position of the pelvis swing to balance the pitch, roll inclinations of the torso Compensate ankle joint trajectories to prevent “til
Predicted	Tilt over controller (stage one and three)	Compensate the ankle joint trajectories and prevent “tilt over” in roll direction
motion control	Landing position and	Compensate landing

	controller (stage two and three)	position to prevent unstable landing

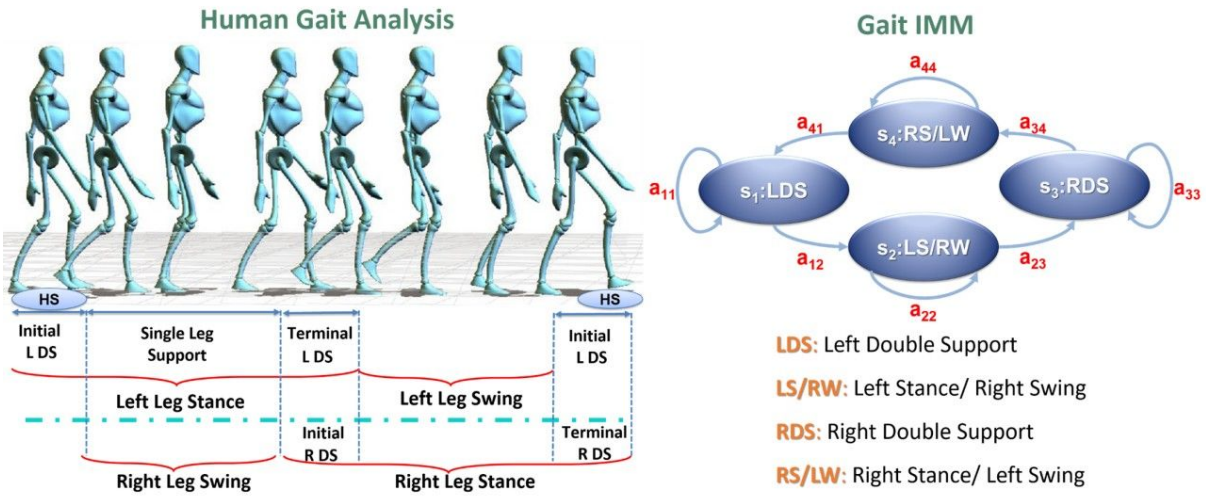
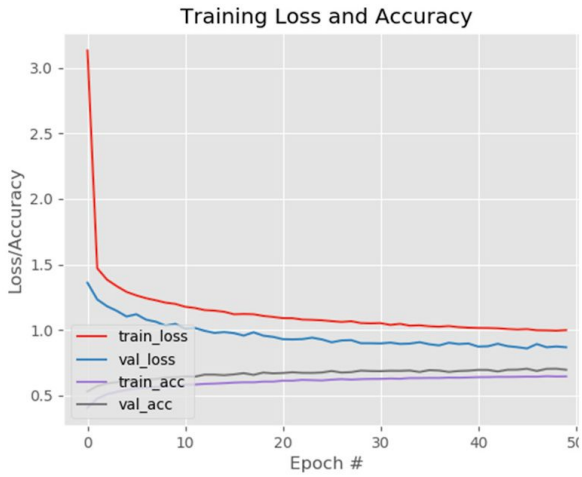


Fig 4.2

CHAPTER 4
RESULT ANALYSIS



(a)

(b)

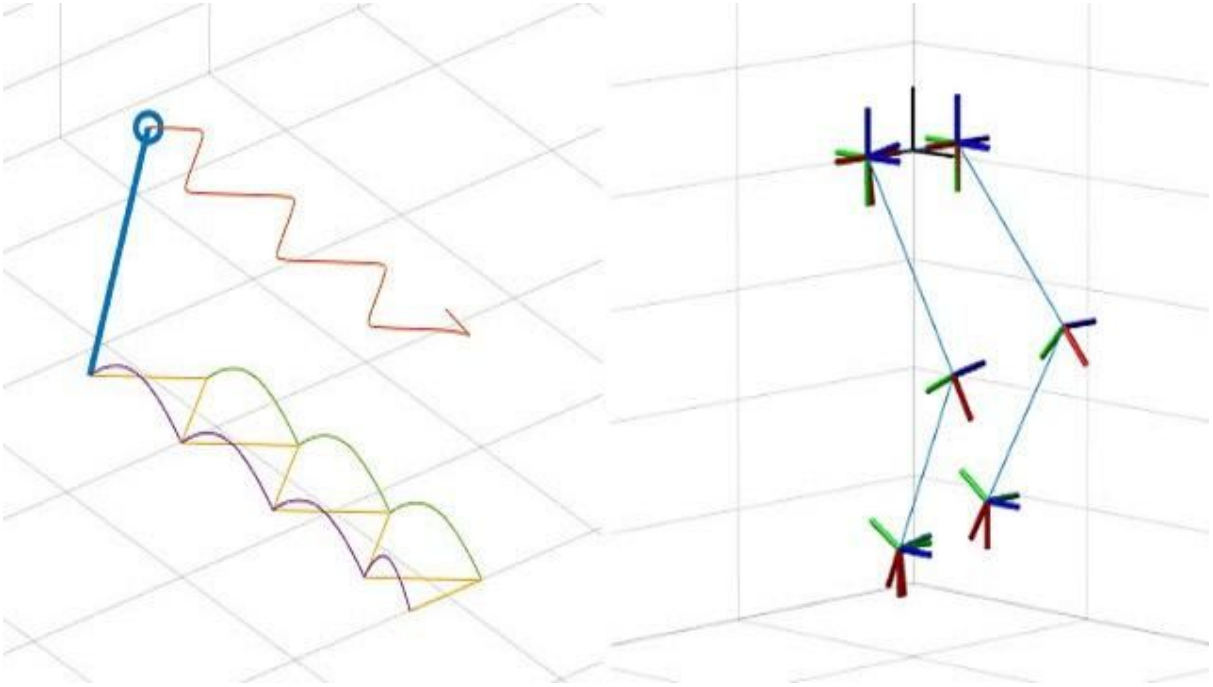
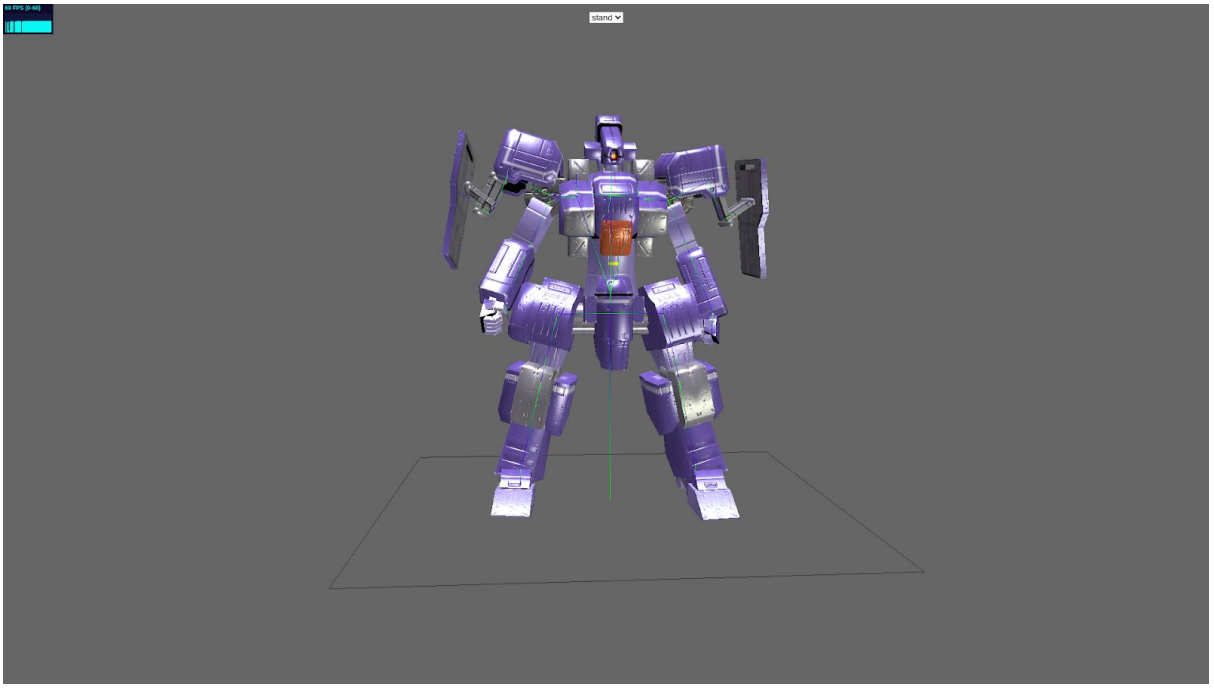


Fig 4.1

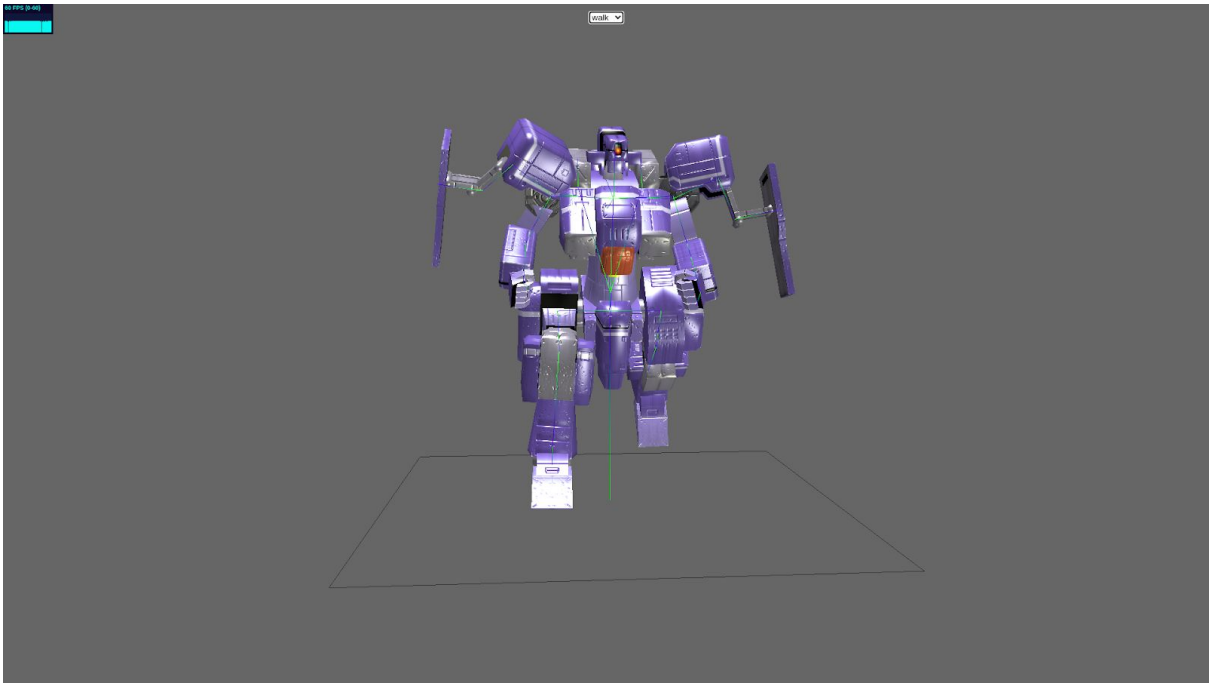
The above shown graph of a left leg is of a graphical mode which shows the maximum angle of the ankle. the left image or graph shows the locus of the path travelled by them.

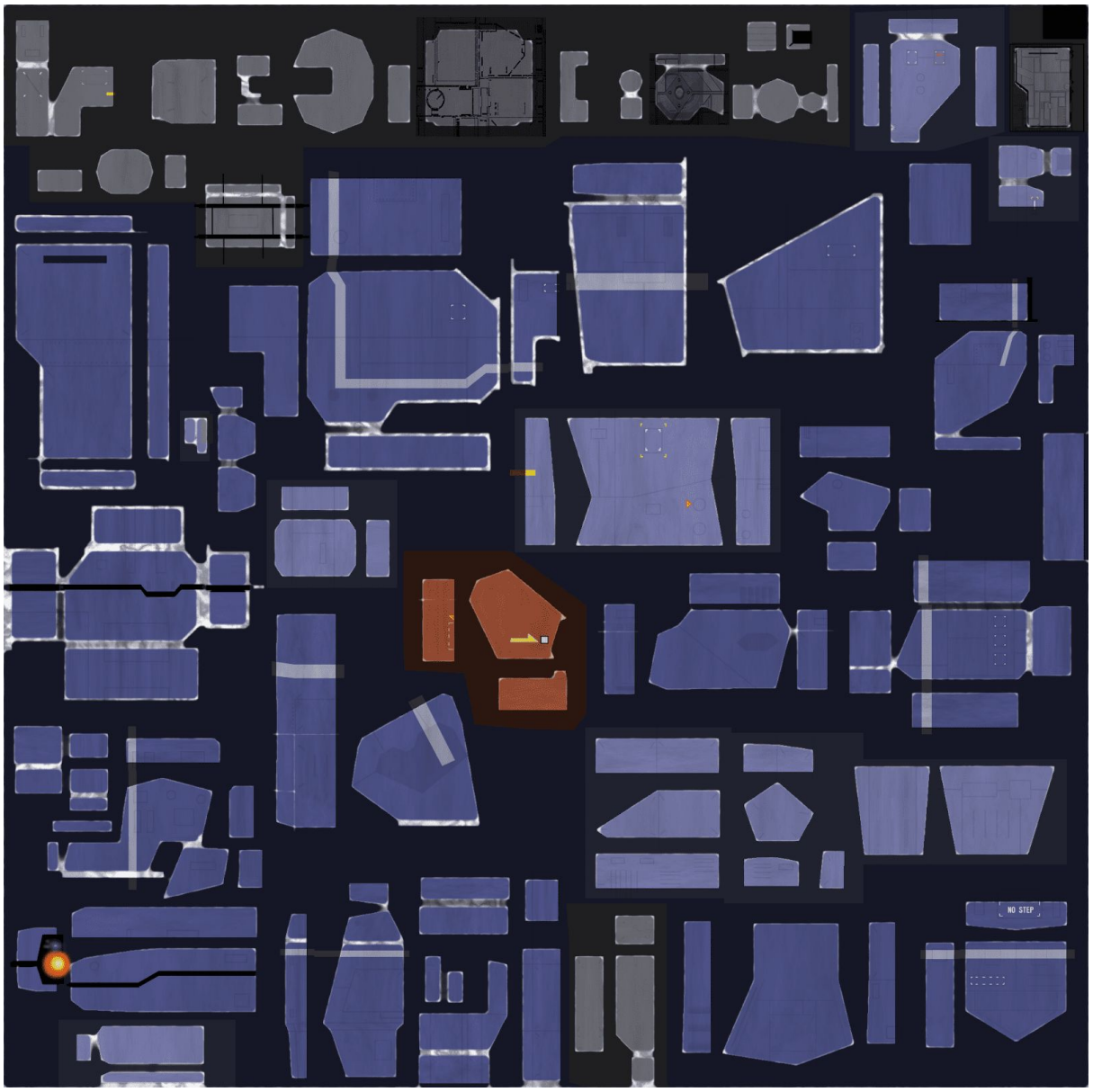
The graph of training and loss is of the DNN due to the gradient descent of the values taken to train, as the training was done to 3:7 ratio. It took 1.2 days approximately to run the complete dialogue box on the cloud, which was helpful.

The model used for testing is shown below. This is a live example of how the Humanoid moves according to the gait system installed here. The third image shows how the 3d file is created individually. When put together, the first image is how it looks. The number of parts created in the third image are the number of nodes, created. But due to similarity or the mirroring phenomenon the only nodes actually needed to be created are 8. All the other nodes are either replicated or ignored as they provide minimal movement providing extra stability mass to the system.



The Blue-Green line is the center of mass of every object, and it moves with respect to the coordinate system on the center of the frame. When you hit the walk button the body walks straight, and with that comes the movement of the center of the mass with it too.


60 FPS (0-60)
walk


NO STEP

CHAPTER 5
CONCLUSION

In conclusion the Artificial intelligence in humanoid robots can be served as much as training data is available. Time is an important factor so the materials where it needs to be precise or the signals entable to mixup and the stability would not attain. The processor used was highly helpful for that reason.

As the latest change here applied here is the compression algorithm, which helps the data transfer time, which reduces the lag by further 8-10%.

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web

You can watch the live simulation/animation on the cloud server hosted on my website that is,

https://the-mstring.co.in

This is our hosted website where you can find the entire source code
it is totally secure as SSL certificate.